

Challenges of Centralized Remote Power Smoothing Strategies for AI Data Center Loads

Abdullah Alfarrar*, Abdallah F. El-Hamalawy *, Ahmed Abd Elaziz Elsayed *, Hany E.Z. Farag*

*EECS Department, York University, Toronto, Ontario, Canada

Abstract—The rapid expansion of the Artificial Intelligence (AI) industry has driven large-scale growth of AI-oriented Data Centers (AIDCs) and their integration into power systems. In addition to their high energy demand, AIDCs rely on accelerator-based infrastructure, including GPUs, TPUs, and NPUs, which exhibit significant power variability due to the synchronized nature of AI training workloads. This synchronized operation produces fast and large power fluctuations that propagate to the regional grid and affect power system frequency stability. Therefore, this paper investigates the impact of ultra-fast varying AIDC loads on power system stability under different power smoothing strategies and highlights the new challenges that distinguish AIDC loads from traditional large industrial demand. Results show that AIDC operation increases frequency deviation and Rate-of-Frequency-Change (ROFC) by 20.15% and 153.29%, respectively, even when the AIDC represents only 40% of a traditional load. Applying local smoothing reduces these increases to 4.43% and 5.26%, respectively, but requires oversizing dedicated local Energy Storage Systems (ESS). In contrast, Centralized Remote Power Smoothing reduces ESS capacity requirements but exhibits a notable performance degradation due to the higher magnitude and frequency of AIDC power fluctuations. These findings demonstrate that conventional smoothing approaches are not fully adequate for the emerging dynamic behavior of AIDC loads.

Index Terms—AI Data Center, AI Training Workload, Power Smoothing, Remote Power Smoothing, Frequency Regulation

I. INTRODUCTION

Recent advances in artificial intelligence (AI) have transformed it from an academic discipline into a trillion-dollar industrial sector, driving rapid expansion of AI-oriented data center (AIDC) infrastructure and associated electricity demand [1]. As a result, global data center (DC) electricity consumption is projected to increase from 415 TWh in 2024 to nearly 945 TWh by 2030, accounting for approximately 10% of global electricity demand growth [2]. Beyond their large energy requirements, AIDCs exhibit unique electrical characteristics that distinguish them from conventional loads. They rely heavily on hardware accelerators such as Graphic Processing Units (GPUs), Tensor Processing Units (TPUs), and Neural Processing Units (NPUs), which operate at extremely high power densities and experience rapid transitions between computational states. These workload-driven transitions produce significant short-term power fluctuations, with ramp rates reaching up to ± 1.9 p.u./sec [3]. When such fast power variability is coupled with the rapidly increasing peak demand of AIDCs, projected to grow from 751 MW in 2025 to 3.3 GW by 2027, these facilities begin to resemble large, highly dynamic industrial loads, which creates a compounded stress on regional power systems [4]. Addressing this emerging challenge requires new operational strategies and power smoothing approaches.

In this regard, existing AI workload power smoothing techniques can be broadly categorized into three groups: 1) software-based methods, 2) GPU-level power ramp control, and 3) hardware-based smoothing using Energy Storage Systems (ESS) [5]. Software-based approaches attempt to introduce virtual or dummy computational loads on accelerator infrastructure during transitions between computational stages. The objective is to reduce abrupt power drops by maintaining a more constant utilization level. However, this strategy often reduces overall training energy efficiency and introduces communication and monitoring overhead due to the additional coordination and kernel-level modifications required in AI workloads. Similarly, GPU-based power ramp constraint methods enforce controlled power transitions at the device level. When GPU power falls below a specified threshold during training or when entering idle states, artificial burn loads are introduced and power is reduced gradually to satisfy ramp-rate limits. Although effective in limiting fast power changes, this approach increases thermal stress, leads to additional power losses, and accelerates hardware aging, thereby shortening GPU lifetime. The third category relies on external hardware-based smoothing using ESS. In this approach, fast charge–discharge cycles buffer the AIDC load, effectively filtering short-term power fluctuations. While technically effective, this solution introduces substantial capital cost and accelerated ESS degradation, primarily due to the extremely high cycling frequency, which can reach millions of cycles per year [5].

Recalling the ESS technologies commonly used for smoothing the power output of Renewable Energy Sources (RESs), several storage options are available, such as battery ESSs, flywheel ESSs, and supercapacitors [6]. Although these technologies have proven effective in smoothing RES power profiles, they require additional capital investment and further considerations when used with AIDCs. Specifically, given the high cycling frequency associated with AIDCs, the use of local ESSs for power smoothing is expected to experience significant degradation, resulting in a substantially reduced lifetime. Further, the large magnitude of power fluctuations increases the likelihood of oversizing local ESSs associated with AIDCs, thereby resulting in higher capital and operational costs.

To reduce the heavy reliance on local ESSs, a recent study [7], conducted by some of the work's authors, proposes a Centralized Remote Power Smoothing (CRPS) configuration in which the smoothing burden of different Fluctuating Power Sources and Loads (FPSLs), including RESs and large variable loads, is addressed collectively. In this configuration, power

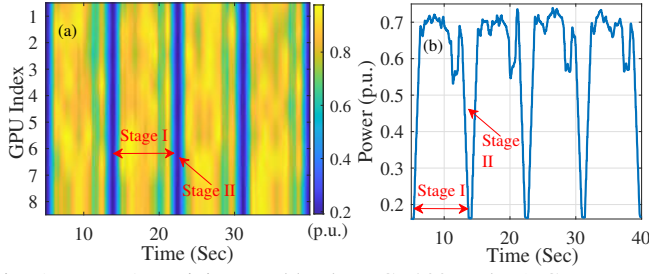


Fig. 1: LLM AI training workload on GB200 node a) GPUs power synchronization within the node, b) Aggregated GB200 node power.

measurements from the FPSLS are transmitted to the regional control center via a high-speed communication network. Then, a central CRPS controller calculates the total required smoothing power and distributes the smoothing burden among multiple Smoothing Facilities (SFs) geographically dispersed across the power grid. The participating SFs include ESSs, fast dispatchable loads, and generating units. The CRPS approach is proposed as an ancillary service that fulfills smoothing requirements by leveraging existing facilities across the power grid, thereby avoiding the need to establish new local infrastructure at each FPSL. Yet, the previous work [7] considered only FPSLS of conventional varying loads and RESs, without addressing the potential of smoothing the ultra-fast power variations of AIDCs using the proposed CRPS approach.

As such, this paper is the first to highlight the differences between smoothing the well-known power fluctuations of RESs and varying conventional loads and the ultra-fast large power variations of AIDCs. Additionally, this work is the first to assess the effectiveness and identify the challenges of the recently proposed CRPS approach in mitigating the AIDCs' ultra-fast power variations.

II. AI DATA CENTER WORKLOADS AND PROBLEM DESCRIPTION

AI training workloads represent a new class of loading conditions in AIDC infrastructure. In addition to their high power densities and intensive reliance on accelerator-based hardware, these workloads operate in a highly synchronized manner. Specifically, large numbers of accelerators transition simultaneously between distinct operational phases, including computation, communication, and information storage stages. This synchronized operation amplifies aggregate power swings at the facility level, as many devices change power states at the same time rather than independently. Fig. 1 illustrates the AI training workload of a Large Language Model (LLM) executed on a GB200 node [8]. As shown in Fig. 1 (a), GPUs within the node operate in a highly synchronized manner, transitioning collectively between high and low power stages, including computational (Stage I) and communication (Stage II) stages. When this synchronized behavior is aggregated at the node level, it produces bursts in power demand and significant short-term power variations over very small time scales, as depicted in Fig. 1 (b).

III. LOCAL VERSUS CENTRALIZED REMOTE POWER SMOOTHING

There are two power smoothing configurations: conventional local power smoothing and the recently proposed CRPS [7]. This section summarizes the two approaches and highlights the differences in their implementation.

A. Conventional Local Power Smoothing

The conventional local power smoothing is typically performed by associating each FPSL with an SF connected to the same bus. In this configuration, assuming there are $\mathcal{N}$ FPSLS, the power output of each FPSL, $P_{FPSL,n}^t$, is smoothed through power injection from its associated SF. The smoothing power setpoint of each SF is computed as the difference between the smoothed power target, $P_{ref,n}^t$, and the FPSL power, such that:

$$P_{SF,n}^{*,t} = P_{ref,n}^t - P_{FPSL,n}^t \quad \forall t \in \mathbb{T} \wedge n \in \mathcal{N}. \quad (1)$$

This setpoint represents a power reference for the SF and is distinct from the actual SF output power $P_{SF,n}^t$, which is subject to dynamic behavior and operational constraints. Consequently, the SF output power may deviate from its setpoint, particularly during transient conditions or when saturation limits are reached. The smoothed power target, $P_n^{*,t}$, can be computed using different methods; one common approach is to calculate it as a moving average over a time window τ , as follows:

$$P_{ref,n}^t = \frac{1}{\tau} \int_{t-\tau}^t P_{FPSL,n}^t dt \quad \forall t \in \mathbb{T} \wedge n \in \mathcal{N}. \quad (2)$$

B. Centralized Remote Power Smoothing

The CRPS is a prospective ancillary service that is supposed to smooth the power fluctuations of different intermittent sources and loads in a collective manner incorporation with the existing Automatic Generation Control (AGC). Unlike conventional local smoothing, in the CRPS, there is no dedicated SF for each FPSL. Yet, the power of all FPSLS is sent via a fast, large-scale communication network to the regional control center, which collectively calculates the total required smoothing power and then distributes the smoothing power to different participating SFs. As such, the CRPS approach can meet smoothing requirements by leveraging existing facilities (e.g., ESSs) across the power grid, thereby avoiding the need to establish new local SFs for each FPSL. Additionally, CRPS benefits from aggregating the power profiles of multiple FPSLS, as the power of one FPSL may increase while another decreases, resulting in lower overall power fluctuations and reduced smoothing requirements.

Fig. 2 shows a portion of a power grid (controlled area), $Area_i$, and the corresponding AGC and CRPS controller. As shown in the figure, the system has $\mathcal{G}$ conventional generators participating in AGC, $\mathcal{N}$ FPSLS, and $\mathcal{M}$ SFs participating in CRPS. The $\mathcal{N}$ FPSLS send their power measurement to the regional control center (i.e., CRPS controller), and the $\mathcal{M}$ SFs receive the smoothing power setpoints from the controller. As indicated in the figure, the CRPS controller acts as a feedforward component that mitigates the power disturbances employed by the FPSLS before being reflected to the system's frequency. Further, in this way, AGC requirements are limited

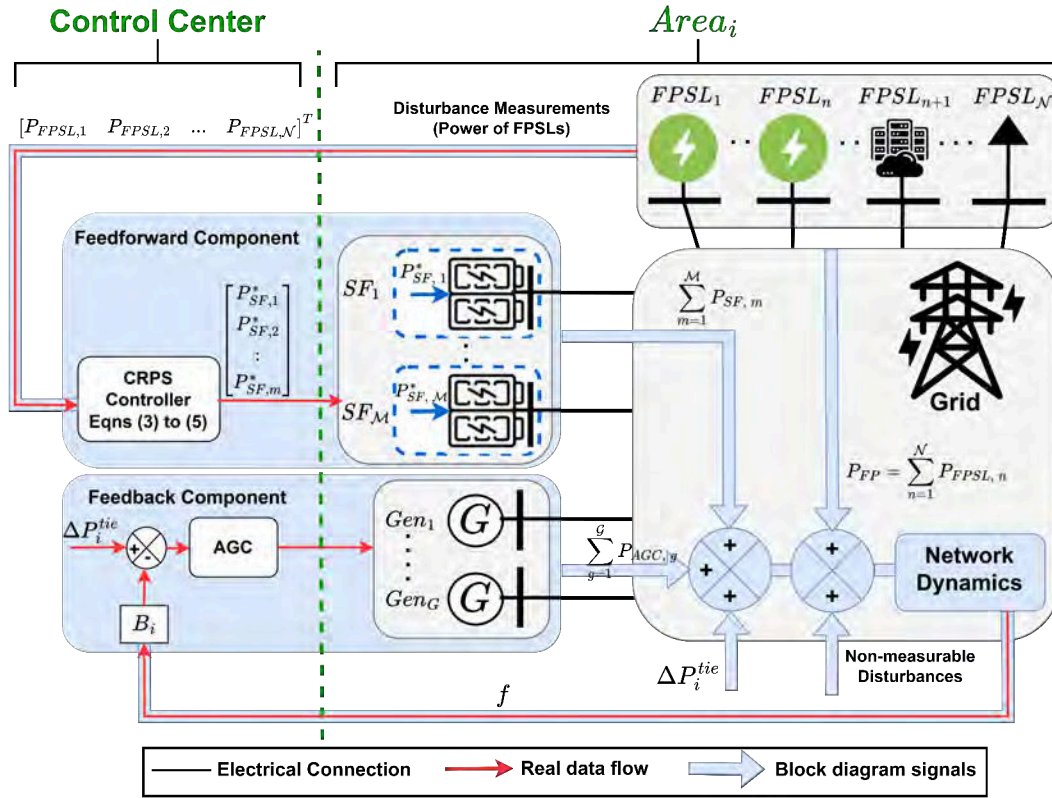


Fig. 2: Architecture of the centralized power smoothing ancillary service.

to other non-measurable disturbances rather than being consumed by the measurable fluctuations of the FPSLs.

The CRPS controller calculates the total fluctuating power injected into the system, P_{FP}^t , by summing the power outputs of the N FPSLs at time t as follows:

$$P_{FP}^t = \sum_{n=1}^N P_{FPSL,n}^t \quad \forall t \in \mathbb{T}. \quad (3)$$

Then the total smoothing power requirement (i.e., smoothing power setpoint), denoted by $P_{CRPS}^{*,t}$, is defined as the difference between the total smoothed power target P_{ref}^t and the total fluctuating power as below:

$$P_{CRPS}^{*,t} = P_{ref}^t - P_{FP}^t \quad \forall t \in \mathbb{T} \quad (4a)$$

where,

$$P_{ref}^t = \frac{1}{\tau} \int_{t-\tau}^t P_{FP}^t dt \quad \forall t \in \mathbb{T}. \quad (4b)$$

Here, τ denotes the smoothing time window in seconds.

The total smoothing power setpoint, $P_{CRPS}^{*,t}$, is then distributed among the M SFs based on their rated assigned power for smoothing. As such, the contribution of each SF m (i.e., $P_{SF,m}^{*,t}$) in $P_{CRPS}^{*,t}$ is calculated as a percentage of its rated assigned power, $P_{SF,m}^{rate}$, allocated for CRPS, relative to the total rated power of all SFs, $\sum_{m=1}^M P_{SF,m}^{rate}$ as follows:

$$P_{SF,m}^{*,t} = \frac{P_{SF,m}^{rate} P_{CRPS}^{*,t}}{\sum_{m=1}^M P_{SF,m}^{rate}} \quad \forall m \in \mathcal{M} \wedge t \in \mathbb{T} \quad (5)$$

It is worth noting that, unlike conventional local smoothing strategies, introducing CRPS as an ancillary service does not require explicitly accounting for the energy limits of participating SFs in the CRPS controller. This is because each SF is

responsible for following the CRPS controller setpoints; if an SF is unable to do so, it should indicate that it is unavailable for participation in CRPS during the given assignment period. Consequently, the grid operator can assign smoothing tasks to other available SFs.

IV. CASE STUDIES AND RESULTS

To evaluate the operational challenges associated with smoothing the high-magnitude ultra-fast power fluctuations characteristic of the AIDCs, six case studies were conducted on the IEEE 39-bus benchmark system shown in Fig. 3. As illustrated in the figure, three FPSLs are connected to buses 16, 32, and 38. The FPSLs at buses 32 and 38 correspond to wind power plants, whereas the FPSL at bus 16 represents a load. In Cases 1, 3, and 5, this load is modeled as a conventional load of 300 MW, whereas in Cases 2, 4, and 6, it is modeled as an AIDC load of 120 MW. Seven SFs are optionally connected to buses 16, 32, 38, 4, 7, 22, and 39, depending on the case study. Specifically, Cases 1 and 2 represent the base scenarios without any power smoothing configuration; therefore, none of the SFs are in operation. Cases 3 and 4 employ conventional local power smoothing, in which only the SFs at buses 16, 32, and 38 are active. In Cases 5 and 6, CRPS is utilized; accordingly, only the remote SFs at buses 4, 7, 22, and 39 are in operation. Table I summarizes the conducted case studies.

The power profiles of the conventional load at bus 16 and the two wind power plants at buses 32 and 38 are based on real, high-resolution power data for a conventional load and wind power plants in Ontario, Canada [7]. The power profile of the AIDC load at bus 16 is modeled as a composite of IT and non-

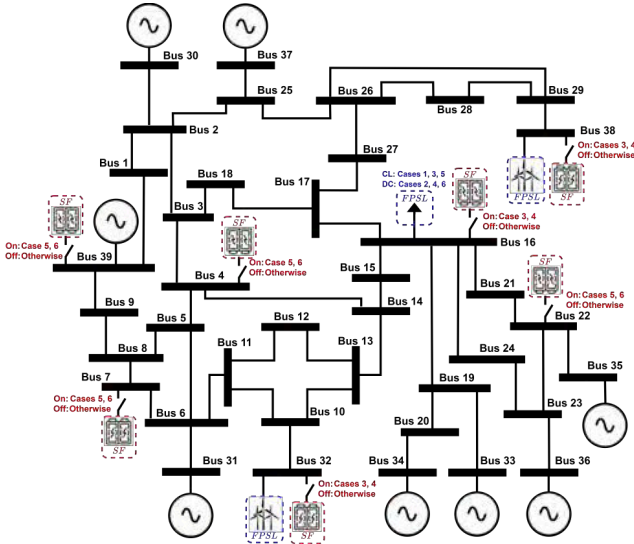


Fig. 3: Single line diagram of the IEEE 39-bus New England benchmark system [9].

TABLE I: Summary of the simulation case studies

Case No.	Type of Load at Bus 16	Smoothing Technique
Case 1	Conventional load	None
Case 2	AIDC load	None
Case 3	Conventional load	Local
Case 4	AIDC load	Local
Case 5	Conventional load	CRPS
Case 6	AIDC load	CRPS

IT loads. The IT load is partitioned into three sectors: i) Cx-A, which represents 56.7% of the AIDC load and is allocated to LLM training workloads; ii) Cx-B, which represents 24% of the AIDC load and corresponds to standard multi-tenant customer workloads and iii) Cx-C, which represents 2.3% of the AIDC load and corresponds to other IT loads. The non-IT load accounts for 17% of the total load and primarily represents the facility's cooling and chiller systems. The LLM power profile (i.e., Cx-A) is adopted from [8] for NVIDIA GB200 node. The power profile for other IT operations (i.e., Cx-B, and Cx-C) and cooling systems was derived from the operational records of a Canadian-based DC solutions provider. The total composition of the DC power is summarized in Fig. 4.

Based on the above load composition, the Power Usage Effectiveness (PUE), defined as the ratio of the total AIDC power consumption to its IT power consumption, is calculated to be approximately 1.2. This value is consistent with industry benchmarks reported by leading providers, such as Schneider Electric [10], indicating that the generated AIDC power profile is realistic and aligned with current AIDC power infrastructure practices. The generated profile is then scaled up to 120 MW, reflecting the actual power demand requirements of high-density AI training facilities [11].

A. Smoothing Performance Measures

This section defines the performance indices and quantitative measures utilized to facilitate a rigorous comparative analysis across the evaluated case studies. To ensure accurate measurement of the smoothing performance, the initial 100

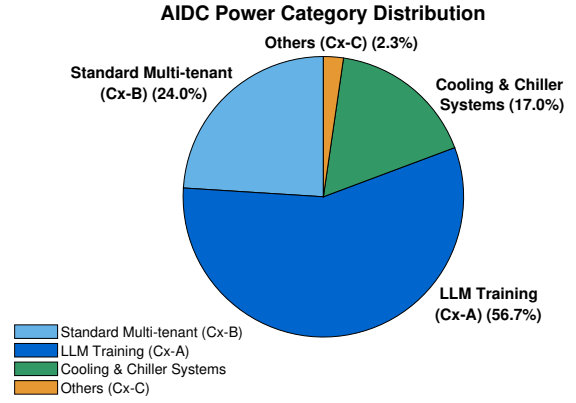


Fig. 4: AIDC Power load distribution summary of the AIDC at bus 16.

seconds of each simulation were excluded to eliminate start-up transients. Also, in this study, the power flow across the tie lines connecting the system to adjacent controlled areas is assumed constant. Consequently, the Area Control Error (ACE) becomes directly proportional to the system frequency deviation ΔF . Hence, two key metrics are utilized to assess the smoothing performance and the effect of the power fluctuations on the electric grid.

1) *Cumulative Absolute Frequency Deviation*: The first metric is the cumulative absolute frequency deviation, $\sum |\Delta F|$, which quantifies the total magnitude of frequency excursions from the nominal steady-state value. It is defined as:

$$\sum |\Delta F| = \sum_{t=t_1}^{\mathbb{T}} |f_n - f(t)|, \quad (6)$$

where $\mathbb{T}$ denotes the total number of time samples in the evaluated period, t_1 is the first time sample after 100 seconds of simulation, $f(t)$ is the system frequency at sample t , and f_n is the nominal steady-state frequency.

2) *Sum of Absolute Rate of Change of Frequency*: The second metric assesses frequency volatility by calculating the sum of the absolute rate of change of the system's frequency $\sum \left| \frac{dF}{dt} \right|$, and it is defined as:

$$\sum \left| \frac{dF}{dt} \right| = \sum_{t=t_1}^{\mathbb{T}} \left| \frac{\nabla f(t)}{\Delta t} \right|, \quad (7)$$

where ∇ denotes the numerical gradient operator and Δt is the simulation sampling interval used for numerical differentiation.

Taken together, these two metrics provide a comprehensive quantitative assessment of the impact of power fluctuations on system stability, while the accompanying time-series figures offer a qualitative visualization that further validates and clarifies the dynamic nature of these frequency variations.

B. Cases 1 and 2: The "AI Stability Penalty"

The initial comparison between the conventional load (Case 1) and the AIDC load (Case 2) underscores the critical stability challenges inherent in integrating AIDC into the grid. Despite the AIDC load representing only 40% of the conventional load's nominal magnitude, its impact on grid stability is disproportionately severe.

TABLE II: System frequency performance comparison for Cases 1–6

Case Study	Smoothing Technique	$\sum \Delta F $ (Hz) ^a	$\sum \frac{dF}{dt} $ (Hz/s) ^a
Case 1: Conventional load	None	221.8480	48.2125
Case 2: AIDC load	None	266.5611	122.1126
Case 3: Conventional load	Local	158.9107	38.8544
Case 4: AIDC load	Local	165.9077	40.2270
Case 5: Conventional load	CRPS	162.2975	39.2837
Case 6: AIDC load	CRPS	183.2945	120.4919

^a The frequency deviations and rate of change are the absolute sum over the simulation period as defined in IV-A.

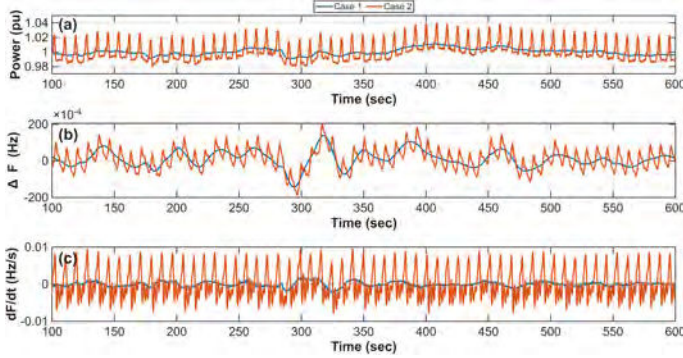


Fig. 5: Comparison of system dynamics between Cases 1 and 2: a) sum of power output of all FPSLS (P_{FP}), b) resultant system frequency deviation, and c) rate of change of frequency.

As illustrated in Fig. 5, the AIDC load exhibits high-frequency, high-amplitude fluctuations that are absent in the conventional load profile. These volatile dynamics exacerbate frequency excursions, as evidenced by the frequency deviation and rate of change of frequency metrics. Quantitatively, Table II confirms that the AI load increases the $\sum |\Delta F|$ and the $\sum |\frac{dF}{dt}|$ by 20.15% and 153.29%, respectively.

C. Cases 3 and 4: AIDC Effect with Local Smoothing

Local smoothing was integrated into the system for both the conventional load and the AIDC load, designated as Case 3 and Case 4, respectively. The objective of this comparison is to characterize the maximum achievable improvement in the AIDC power profile relative to a conventional load.

The resultant smoothed power profiles are shown in Fig. 6. Although the power profiles in Cases 3 and 4 appear effectively smoothed compared with Cases 1 and 2, respectively, Fig. 7 (a) shows that the net smoothed power in Case 4 exhibits notable high-frequency, low-amplitude ripples that are absent in Case 3. Further, as shown in Fig. 7 (b) and (c), the frequency deviation and rate of change under AIDC loading (Case 4) display similar low-amplitude ripples, indicating that the AIDC's high-frequency components are not fully suppressed.

Quantitative metrics detailed in Table II support this observation; the $\sum |\Delta F|$ and the $\sum |\frac{dF}{dt}|$ for the AIDC are greater than those of the conventional load by 4.43% and 5.26%, respectively.

These results highlight that, despite the high investments required for local smoothing of AIDCs discussed earlier in section I, the inherent volatility of AIDC demand leaves a

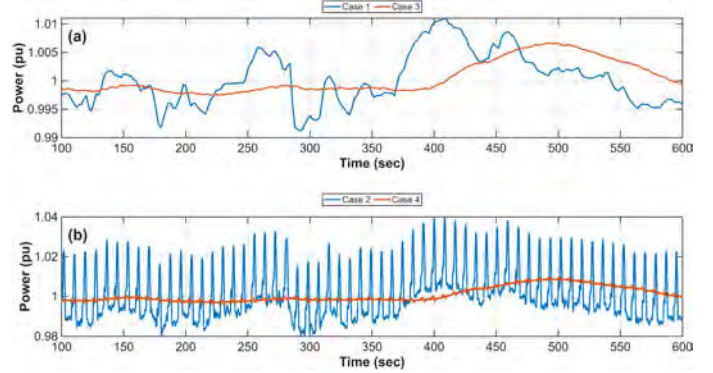


Fig. 6: Comparison of the total smoothed power output of all FPSLS and SFs (i.e., $P_{FP} + \sum_{n=1}^N P_{SF,n}$) for: a) Cases 1 and 3, and b) Cases 2 and 4.

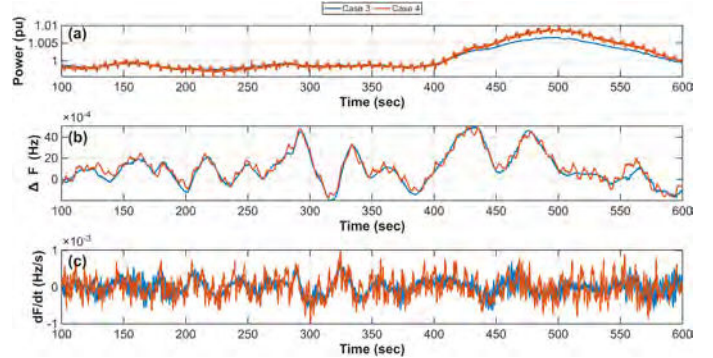


Fig. 7: Comparison of system dynamics between Cases 3 and 4: a) net smoothed power output of all FPSLS and SFs (i.e., $P_{FP} + \sum_{n=1}^N P_{SF,n}$), b) resultant system frequency deviation, and c) rate of change of frequency.

spectral footprint on the grid frequency that is more difficult to suppress than that of conventional loads and RESs.

D. Cases 5 and 6: AIDC Effect with Centralized Smoothing

The CRPS approach recently proposed by [7] is applied in Cases 5 and 6, replacing the conventional local smoothing used in Cases 3 and 4. While [7] reported negligible differences between local smoothing and CRPS, the integration of AIDC load introduces substantially higher volatility than previously anticipated. As illustrated in Fig. 8, although the power appears effectively smoothed in subplot (a), the system frequency deviation and rate of change in subplots (b) and (c) exhibit substantial fluctuations.

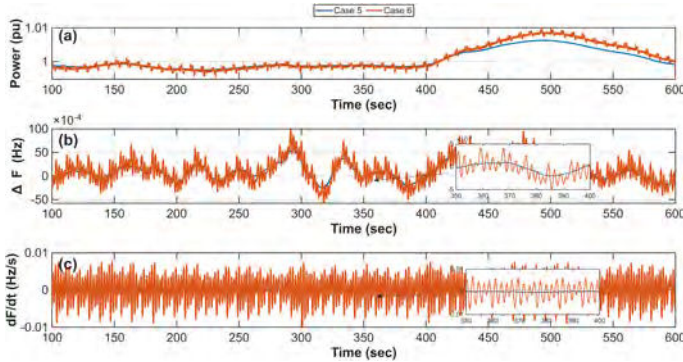


Fig. 8: Comparison of system dynamics between Case 5 and Case 6: a) net smoothed power output of all FPSLs and SFs (i.e., $P_{FP} + \sum_{m=1}^M P_{SF,m}$) b) resultant system frequency deviation, and c) rate of change of frequency.

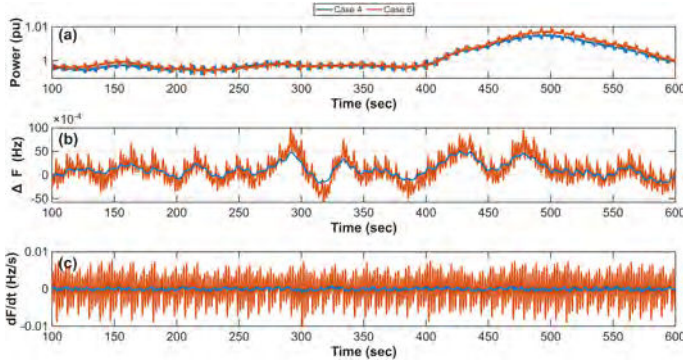


Fig. 9: System performance comparison for Cases 4 and 6: a) net smoothed power output of all FPSLs and SFs, b) resultant system frequency deviation, and c) rate of change of frequency.

This observed discrepancy between the smoothed power profiles and the corresponding system frequency fluctuations can be attributed to the different losses experienced by the power injected from FPSLs and that injected by the SFs, as well as to the geographical displacement of remote SFs, which may introduce minor power propagation delays. Although these effects are present in the case of RESs and conventional loads (i.e., Case 5), they become more pronounced under AIDC loading (i.e., Case 6) due to the significantly higher magnitude and frequency of power fluctuations compared with scenarios involving only RESs and conventional load variations. Quantitative metrics further substantiate these observations: the $\sum |\Delta F|$ and the $\sum |\frac{dF}{dt}|$ for the AIDC case (i.e., Case 6) are higher than those of Case 5 by 12.96% and 207.69%, respectively, as detailed in Table II.

While the results from Cases 3 and 4 confirmed that AIDC load degrades power quality when it is smoothed locally on-site, the performance of CRPS in Case 6 was further deteriorated when compared by Case 4 as shown in Fig. 9.

These findings underscore the critical nature of AIDC smoothing and serve as a critical warning regarding the limitations of smoothing architectures in general and centralized smoothing (i.e., CRPS) specifically. Despite the benefits of the CRPS approach as a prospective ancillary service for reducing AGC energy requirements and needed SFs infrastructure, as noted in [7], its current form may be insufficient to address

the ultra-fast and high-magnitude power fluctuations associated with AIDC loads.

V. CONCLUSION

This paper evaluated the grid stability impacts of the ultra-fast, high-magnitude power variations introduced by AIDCs dominated by NVIDIA GB200 nodes during LLM training. Simulation studies on the IEEE 39-bus system confirmed that AI training workloads exhibit a distinctly more volatile electrical signature than conventional industrial loads. Results showed that although on-site local smoothing techniques improve the AIDC power profile, their performance remains inferior to that of traditional industrial demand, indicating that existing smoothing approaches are insufficient to address the unique dynamics of AI training loads. Furthermore, the CRPS framework demonstrated limited effectiveness in mitigating these ultra-fast transients, with the frequency rate of change improving by only 1.33% compared to the unsmoothed case, revealing a clear technical limitation of current centralized strategies. Overall, the findings provide a critical warning: as high-density AIDC training continues to scale, both traditional and centralized smoothing architectures must be fundamentally enhanced to preserve regional grid stability.

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